ROCURONIUM

High-alert drug

Reversed by sugammadex

CHEMICAL NAME 1-[(2β,3α,5α,16β,17β)-17-acetoxy-3-hydroxy-2-(morpholin-4-yl)androstan-16-yl]-1-allylpyrrolidinium bromide

CLASS Non-depolarising neuromuscular blocker (aminosteroid), intermediate-acting

TRADE NAMES Esmeron · Zemuron · Rocuronium bromide injection

THERAPEUTIC USES / INDICATIONS

- Intubation, including rapid sequence induction (high dose) when suxamethonium is contraindicated.
- Muscle relaxation during surgery.
- ICU mechanical ventilation.

ADMINISTRATION

- **IV** bolus or infusion.
- Store at 2–8 °C (60 days at room temperature).
- Patient must be sedated and ventilated.
- May cause pain on injection in awake patient.

DOSAGES

- **Intubation:** 0.6 mg/kg.
- **Rapid sequence induction:** 1–1.2 mg/kg.
- **Maintenance:** 0.15 mg/kg or infusion 0.3–0.6 mg/kg/h.
- Reduce in elderly, liver disease.

PHARMACOKINETICS

- **Onset:** 60–90 s (high dose); 2 min (standard). **Duration:** 30–60 min.
- Mainly hepatic uptake and biliary excretion; some renal.
- Prolonged in liver failure. **t½:** 1–2 h.

PHARMACODYNAMICS

- Competitive antagonist at **nicotinic ACh receptors** → muscle paralysis.
- No histamine release; minimal cardiovascular effects.
- Specifically encapsulated by sugammadex (rapid reversal) or reversed by neostigmine.

SIDE & ADVERSE EFFECTS

SIDE EFFECTS (COMMON)

- Injection pain; mild tachycardia.
- Transient hypotension or hypertension.

ADVERSE EFFECTS (SERIOUS)

- Anaphylaxis (one of commoner causes in anaesthesia).
- Residual paralysis post-operatively.
- Awareness if sedation inadequate.
- Prolonged weakness in ICU.

CONTRAINDICATIONS & CAUTIONS

- Hypersensitivity to rocuronium or bromide.
- **Caution:** liver disease, myasthenia gravis, burns (resistance), obesity (dose by ideal weight).

KEY INTERACTIONS

- **Volatile anaesthetics, aminoglycosides, magnesium:** prolong block.
- **Anticonvulsants (chronic):** shorten block.

NURSING MANAGEMENT

- Confirm sedation before giving.
- Monitor TOF, SpO_2, ventilation.
- Ensure sugammadex or neostigmine + glycopyrrolate available.
- Post-op: assess for residual weakness (head lift, grip, breathing).
- ICU: eye and skin care.

PATIENT EDUCATION

- Report any difficulty swallowing or breathing after surgery.
- Explain to families that paralysed patients may still hear.

KEY POINTS

- Aminosteroid non-depolarising blocker.
- High dose (1.2 mg/kg) for rapid sequence induction.
- Reversed rapidly by sugammadex.
- Hepatic/biliary elimination.

SOURCES CHECKED **[93]** Rocuronium 10 mg/ml solution for injection / infusion (UK SmPC) **[94]** Rocuronium, IV (US label) **[95]** Rocuronium, IV (US label, Medical Purchasing Solutions)

Full references p. 432–457